Semantic Reliability and Exposure-Conditioned Consensus Estimation in Ranked Social Discussion Streams: Sensitivity, Endogeneity, and Empirical Auditing

Syed Saifullah S and Harish S

***Abstract*—Automated sentiment aggregation over ranked social media discussion feeds (e.g., YouTube comment sections) is widely used to evaluate collective reception regarding public events, media releases, and consumer products. The reliability of such estimates is governed by two independent layers. The semantic layer must map informal, idiomatic, and meme-laden text to a polarity valence; the exposure layer determines which comments are seen, because early-arriving and algorithmically prioritized comments accumulate visibility and endorsement independently of broad community agreement. Standard pipelines typically trust both layers without empirical auditing.**

In this paper we audit both layers and introduce an exposure-conditioned estimator. On the semantic layer, we manually annotate a dated, API-provenanced YouTube corpus ($N = 100$, `dQw4w9WgXcQ`) and show that the widely used VADER lexicon attains only 65.0% accuracy and 0.5307 macro-F1, with 46% zero-inflation, a 23.9% rate of positive comments misclassified as negative, and a 12.5% negative-class precision. On the exposure layer, we formulate PACE (Preferential-Attachment Corrected Endorsement), which models expected engagement as an empirical function of display rank (r_i) and arrival delay (Δt_i), applies Duan's smearing retransformation to correct log-linear retransformation bias, and isolates the Exposure-Conditioned Engagement Residual ($\delta_i = L_i - \hat{\mathbb{E}}[L_i \mid r_i, \Delta t_i]$) through robust sigmoidal gating.

Through deterministic Monte Carlo simulations ($M = 50$ trials per scenario), component ablations, and systematic sensitivity audits, we show that consensus weighting is highly sensitive to estimand definitions and baseline specifications: PACE reduces MAE relative to heuristic weighting in pre-specified settings (e.g., 0.010 ± 0.001 vs. 0.107 ± 0.001 when targeting a late-emerging groundswell), but its advantage vanishes or reverses when targeting overall participant means or symmetric consensus. On real data, all five estimators remain statistically indistinguishable (every point estimate in 0.079–0.123 falls within PACE's 95% bootstrap interval $[-0.03, 0.17]$; permutation $p \geq 0.21$). We conclude that both the semantic and exposure layers must be treated as auditable diagnostics rather than trusted components, and release our complete audit suite and data for reproducibility.

***Index Terms*—Position Bias, Temporal Exposure Decay, Sentiment Aggregation, Lexicon Reliability, Semantic Analysis, Robust Estimation, Social Discussion Streams.**

I. Introduction

ANALYZING collective public sentiment from online comment feeds is a foundational task in computational social science, media analytics, and digital governance [1], [2]. Standard sentiment aggregation pipelines assign each comment c_i a polarity valence $s_i \in [-1.0, +1.0]$ via a natural language processing (NLP) classifier (e.g., VADER [3] or SentiStrength [4]) and compute the arithmetic sample average:

$$\hat{\mu}_{\text{raw}} = \frac{1}{N} \sum_{i=1}^{N} s_i \tag{1}$$

While computationally trivial, arithmetic averaging treats every text identically, ignoring the community's endorsement or rejection of that text. A zero-like unverified assertion carries identical weight to a comment endorsed by thousands of peers [2].

To incorporate community feedback, academic literature and commercial analytics platforms frequently employ **popularity-weighted heuristics**:

$$w_i = 1 + \ln(1 + \text{likes}_i) + 0.5 \ln(1 + R_i) \tag{2}$$

where R_i is the number of replies to comment c_i. This is the Heuristic Log-Likes baseline evaluated throughout this paper.

However, observational upvote counts on modern social platforms are **not** clean measures of text quality or community agreement [5], [6]. Instead, they are confounded by two structural factors:

1) **Position Bias:** Comments displayed at top visual ranks receive orders of magnitude more impressions than lower-ranked comments [7]–[9].
2) **Temporal Precedence:** Comments posted immediately after content publication accumulate engagement over longer durations and capture early viewer traffic [10]–[12].

Consequently, **heuristic like-weighting inherently amplifies exposure bias**. A polarizing or low-effort comment posted within minutes of a video's release can accumulate thousands of upvotes purely due to top-ranked visibility, skewing the aggregate sentiment score [5], [6]. Conversely, an insightful critique posted hours later at rank #50 receives minimal exposure and is discounted by like-weighted aggregators.

A. Scope and Target Estimand

We define our target estimand as the **Engagement-Adjusted Consensus among Observable Discourse Participants**

Syed Saifullah S and Harish S are with the School of Computer Science Engineering and Information Systems (SCORE), Vellore Institute of Technology (VIT), Vellore, Tamil Nadu 632014, India (e-mail: syedsaifullah.s2024@vitstudent.ac.in). Corresponding author: Syed Saifullah S.

($\mu_{\text{discourse}}$). We explicitly do **not** claim to recover the opinions of the "silent majority" (viewers who never comment or upvote), which requires external representative polling [13], [14]. Furthermore, because platforms do not expose individual user impression logs or client-side dwell times to public APIs, our framework performs **exposure-conditioned adjustment** using observable rank and arrival proxies, rather than formal causal identification [15], [16].

B. Key Contributions

The main contributions of this work are summarized as follows:

- **Semantic Reliability Audit of Rule-Based Lexicons:** We annotate all 100 comments of a dated, API-provenanced YouTube snapshot and quantify VADER's degradation on informal discourse, reporting per-class precision/recall/F_1 (Table III), 65.0% overall accuracy, 0.5307 macro-F1, 46% zero-inflation, and a 23.9% positive-to-negative misclassification rate driven by slang, idiomatic negation, and lyric quotation.
- **Exposure-Conditioned Baseline Model:** We formalize expected comment engagement $\mathbb{E}[L_i \mid r_i, \Delta t_i]$ via a closed-form log-linear regression combined with Duan's smearing retransformation factor [17] to correct for retransformation bias under stated error assumptions.
- **Exposure-Conditioned Engagement Residuals & Weight Auditing:** We define the residual δ_i to quantify whether a comment overperformed or underperformed relative to its exposure opportunity. Rather than asserting theoretical non-domination, we evaluate weight concentration empirically via the Maximum Normalized Weight Share ($w^*_{\max} = w_{\max}/\sum w_i$) and the Effective Sample Size (ESS).
- **Replicated Multi-Trial Validation ($M = 50$) and Sensitivity Analysis:** We evaluate PACE against three established baselines across $M = 50$ independent simulation trials per scenario, reporting Paired Wilcoxon Signed-Rank Tests [18] on Mean Absolute Error (MAE), top-10% pruning stability, a component ablation study, and an open ground-truth sensitivity analysis mapping performance across diverse target specifications.
- **Observational Case Study & Diagnostic Audit:** We apply the method to dated YouTube snapshots across four domains ($N = 100$ each) with SHA-256 provenance, reporting descriptive weight diagnostics and systematically auditing practical constraints including ranking endogeneity, rule-based lexicon zero-inflation, and baseline scale sensitivity.

II. Related Work

A. Position Bias and Counterfactual Ranking

Position bias in list-based user interfaces is well-established in Information Retrieval (IR) [9]. Joachims et al. [7] formalized the examination hypothesis: the probability of interaction is a product of visual examination propensity and intrinsic relevance. Inverse Propensity Scoring (IPS) and Normalized Clipped IPS (NC-IPS) weight observed interactions inversely by examination probability [8], [19]. Vardasbi et al. [20] analyzed conditions under which position bias estimation remains identifiable. While extensively studied in search engines and recommender systems, IPS frameworks have historically focused on item ranking rather than continuous opinion valence aggregation across temporal discussion threads.

B. Conversation Trees and Social Dynamics

Modeling social media discussions as hierarchical trees dates back to foundational work by Zubiaga et al. [21], who modeled conversation trees for rumour stance classification (support, denial, query, comment). Gómez et al. [22] characterized structural and temporal branching patterns in threaded discussions. Recent studies in Springer's *Social Network Analysis and Mining* (SNAM) have examined algorithmic drift in YouTube recommendations [23] and anomalous commenter behavior [24]. Our framework does not claim to perform semantic tree stance classification; rather, we incorporate a structural tree damping factor to reflect the narrower conversational scope and lower visual reach of nested reply chains.

C. Self-Selection and the Spiral of Silence

The "spiral of silence" theory introduced by Noelle-Neumann [14] posits that individuals perceiving their opinion to be in the minority tend to remain silent. Zhao, Yang, and Bernstein [13] empirically established that up to 72.1% of online community members perceiving themselves in the ideological minority refrain from public posting. This self-selection confirms that raw text counts represent vocal participants rather than whole viewer populations, reinforcing the necessity of carefully defining the target estimand as discourse consensus.

D. Robust Estimation in Social Feedback

Social interaction distributions exhibit heavy tails and vulnerability to strategic manipulation [5], [12]. Classic robust statistics provide M-estimators and bounded-influence influence functions to limit the damage of arbitrary contamination [25]–[27]. PACE builds upon these principles by standardizing residuals via the Median Absolute Deviation (MAD) and passing them through a sigmoidal transfer function. For aspect-level and fine-grained sentiment benchmarks, the SemEval shared tasks [28] have established evaluation protocols that inform how sentiment classifiers are assessed.

E. Rule-Based Lexicons versus Neural Sentiment Models

Rule-based lexicons such as VADER [3] and SentiStrength [4] remain widespread for social media sentiment because they are deterministic, dependency-light, and interpretable. They are, however, known to degrade on informal text, where negations, slang, emoticons, irony, and dialect shift the polarity signal in ways that hand-crafted lexicons capture poorly [1]. Pretrained neural models—including BERT [29] and Twitter-adapted variants trained on social media corpora [30]—attain markedly higher accuracy on informal sentiment benchmarks, at the cost of compute, opacity, and their own distributional sensitivity. Because our goal is to audit the components

Algorithm 1 PACE: Preferential-Attachment Corrected Endorsement

Require: Comment stream $\mathcal{C} = \{c_i = \langle s_i, L_i, r_i, \Delta t_i, d_i \rangle\}_{i=1}^{N}$, tree damping factor $\phi \in (0, 1]$
Ensure: Consensus estimate $\hat{\mu}_{\text{PACE}}$, normalized weight vector $\mathbf{w}^*$

Formulate design matrix $\mathbf{X} \in \mathbb{R}^{N \times 3}$ with rows $\mathbf{x}_i = [1, \ln(r_i), \ln(1 + \Delta t_i)]$ and target $\mathbf{y} \in \mathbb{R}^N$ with $y_i = \ln(L_i + 1)$
Estimate exposure surface parameters via OLS: $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$
Compute OLS residuals: $e_j = y_j - \mathbf{x}_j^T\hat{\boldsymbol{\beta}}$ for $j = 1, \ldots, N$
Calculate Duan smearing retransformation factor: $s_{\text{smear}} = \frac{1}{N}\sum_{j=1}^{N} \exp(e_j)$
for each comment $i = 1, \ldots, N$ **do**
    Compute expected baseline engagement: $\hat{\mathbb{E}}[L_i \mid r_i, \Delta t_i] = \max\left(0, s_{\text{smear}} \exp(\mathbf{x}_i^T\hat{\boldsymbol{\beta}}) - 1\right)$
    Isolate engagement residual: $\delta_i = L_i - \hat{\mathbb{E}}[L_i \mid r_i, \Delta t_i]$
end for
Standardize residuals: $z_i = (\delta_i - \text{Median}(\boldsymbol{\delta}))/(1.4826 \cdot \text{MAD}(\boldsymbol{\delta}) + \epsilon)$
for each comment $i = 1, \ldots, N$ **do**
    Apply sigmoidal gating and tree damping: $w_i = \sigma(\text{clip}(z_i, -20, 20)) \cdot \phi^{d_i}$
end for
Normalize weights: $w_i^* = w_i / \sum_{j=1}^{N} w_j$
Compute consensus score: $\hat{\mu}_{\text{PACE}} = \sum_{i=1}^{N} w_i^* s_i$
return $\hat{\mu}_{\text{PACE}}$, $\mathbf{w}^*$

of a consensus pipeline rather than to maximize classifier accuracy, we deliberately retain VADER and quantify its failure modes on an out-of-domain, meme-dominated corpus; the audit methodology itself applies identically to any classifier that maps text to a polarity valence.

III. Proposed Methodology: PACE

Let a discussion stream be represented by N comments:

$$c_i = \langle X_i, s_i, L_i, r_i, \Delta t_i, d_i \rangle, \quad i = 1, \ldots, N \tag{3}$$

where X_i is the text, $s_i \in [-1.0, +1.0]$ is the sentiment valence, $L_i \in \mathbb{N}_0$ is the upvote count, $r_i \in \mathbb{N}^+$ is display rank, $\Delta t_i \geq 0$ is the elapsed hours between media publication and comment submission, and $d_i \in \{0, 1, 2, \ldots\}$ is the hierarchical reply depth.

A. Closed-Form Baseline Engagement Estimation

Attention on social media decays power-law with display rank ($r^{-\gamma}$) and heavy-tailed with arrival delay ($(1 + \Delta t)^{-\lambda}$) [9], [11]. We model log-transformed counts:

$$\ln(L_i + 1) = \beta_0 + \beta_1 \ln(r_i) + \beta_2 \ln(1 + \Delta t_i) + \varepsilon_i \tag{4}$$

Let $\mathbf{X} \in \mathbb{R}^{N \times 3}$ have row vectors $\mathbf{x}_i = [1, \ln(r_i), \ln(1 + \Delta t_i)]$, and $\mathbf{y} \in \mathbb{R}^N$ have elements $y_i = \ln(L_i + 1)$. The parameter vector $\hat{\boldsymbol{\beta}} = [\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2]^T$ is solved analytically via Ordinary Least Squares:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} \tag{5}$$

B. Duan's Smearing Retransformation Correction

By Jensen's inequality, naively exponentiating log predictions underestimates the arithmetic conditional mean: $\exp(\mathbb{E}[\ln(L + 1)]) \leq \mathbb{E}[L + 1]$. Under the assumption that residuals ε_i are independent and identically distributed with constant variance, we apply **Duan's (1983) smearing factor** [17]:

$$s_{\text{smear}} = \frac{1}{N}\sum_{j=1}^{N} \exp(e_j), \quad e_j = y_j - \mathbf{x}_j^T\hat{\boldsymbol{\beta}} \tag{6}$$

The calibrated expected baseline engagement is computed as:

$$\hat{\mathbb{E}}[L_i \mid r_i, \Delta t_i] = \max\left(0.0,\ s_{\text{smear}} \cdot \exp(\mathbf{x}_i^T\hat{\boldsymbol{\beta}}) - 1.0\right) \tag{7}$$

C. Exposure-Conditioned Engagement Residuals (δ_i)

The residual isolates excess or deficient engagement relative to the expected exposure baseline:

$$\delta_i = L_i - \hat{\mathbb{E}}[L_i \mid r_i, \Delta t_i] \tag{8}$$

- If $\delta_i < 0$: An early, highly visible comment received fewer likes than its placement predicted.
- If $\delta_i > 0$: A buried or late-arriving comment accumulated likes in excess of its exposure baseline.

D. Sigmoidal Gating and Weight Concentration Analysis

Because social interaction distributions exhibit heavy tails, sample variance is sensitive to extreme viral spikes. We standardize residuals using the robust Median Absolute Deviation (MAD) [25], [27]:

$$z_i = \frac{\delta_i - \text{Median}(\boldsymbol{\delta})}{1.4826 \cdot \text{MAD}(\boldsymbol{\delta}) + \epsilon} \tag{9}$$

Standardized residuals are passed through a logistic sigmoid clipped to $[-20, +20]$ to yield unnormalized weights:

$$w_i^{\text{PACE}} = \frac{1}{1 + \exp(-z_i)} \cdot \phi^{d_i} \tag{10}$$

where $\phi \in (0, 1]$ (default $\phi = 0.8$) is an empirical structural tree hierarchy damping factor for replies at depth $d_i \geq 1$.

The final consensus estimate is the normalized ratio:

$$\hat{\mu}_{\text{PACE}} = \frac{\sum_{i=1}^{N} w_i^{\text{PACE}} s_i}{\sum_{i=1}^{N} w_i^{\text{PACE}}} \tag{11}$$

Remark on Weight Domination: While sigmoidal activation maps unnormalized weights into $(0, 1)$, it does not mathematically bound the normalized weight share:

$$w_{\max}^* = \frac{\max_i w_i}{\sum_{j=1}^{N} w_j} \tag{12}$$

If one comment receives $w_1 \to 1$ while all other $w_i \to 0$, $w_{\max}^* \to 1.0$. We therefore audit weight concentration empirically via $w_{\max}^*$ and the Effective Sample Size (ESS) rather than asserting theoretical non-domination. The complete estimation procedure is formalized in Algorithm 1.

IV. Experimental Protocol and Generative Specifications

A. Synthetic Scenario Generative Specifications

To ensure complete reproducibility, the generative processes for all synthetic scenarios are defined below:

1) **Scenario 1 (Viral Outlier Attack, $N = 50$, Ground Truth $Y^* = 0.600$):**
   - Comment c_0: $s_0 = -0.90$, $L_0 = 15,000$, $R_0 = 250$, $r_0 = 1$, $\Delta t_0 = 0.10$h.
   - Comments $c_1 \dots c_{49}$: $s_i \sim \mathcal{N}(0.60, 0.05^2)$ clipped to $[-1, 1]$, $L_i \sim \text{Uniform}(5, 50)$, $R_i \sim \text{Uniform}(0, 3)$, $r_i = i + 1$, $\Delta t_i = 0.5 + 0.4i$h.
2) **Scenario 2 (Late Organic Groundswell, $N = 60$, Ground Truth $Y^* = -0.100$):**
   - Early hype $c_0 \dots c_9$: $s_i \sim \mathcal{N}(0.70, 0.05^2)$, $L_i \sim \text{Uniform}(150, 350)$, $r_i = i + 1$, $\Delta t_i = 0.1 + 0.2i$h.
   - Casual comments $c_{10} \dots c_{33}$: $s_i = 0.10$, $L_i \sim \text{Uniform}(10, 40)$, $r_i = i + 1$, $\Delta t_i = 2.0 + 0.5i$h.
   - Critical critique c_{34}: $s_{34} = -0.85$, $L_{34} = 800$, $R_{34} = 45$, $r_{34} = 35$, $\Delta t_{34} = 24.0$h.
   - Agreeing feedback $c_{35} \dots c_{59}$: $s_i \sim \mathcal{N}(-0.40, 0.05^2)$, $L_i \sim \text{Uniform}(20, 60)$, $r_i = i+1$, $\Delta t_i = 25.0 + 0.3(i - 35)$h.
3) **Scenario 3 (Democratic Balanced Consensus, $N = 50$, Ground Truth $Y^* = 0.000$):**
   - 25 positive ($+0.50$) and 25 negative (-0.50) comments permuted independently of rank.
   - Expected baseline likes: $\mathbb{E}[L_i] = \lfloor 1000 \cdot r_i^{-0.7} \cdot (1 + 0.3i)^{-0.2} \rfloor + \epsilon_i$, where $\epsilon_i \sim \text{Uniform}(-15, 15)$.

Disclosure of Data-Generation Mechanics: A critical methodological distinction exists across these generators. In Scenarios 1 and 2, likes are drawn from piecewise uniform distributions across comment cohorts or assigned as fixed constants, rather than generated from the continuous parametric rank-and-time exposure surface assumed by PACE. These scenarios evaluate how estimators behave under localized engagement shocks (e.g., a viral outlier or an abrupt late surge) when the data deviates from parametric assumptions. In contrast, Scenario 3 explicitly generates likes from a parametric exposure model $\mathbb{E}[L_i] \propto r_i^{-0.7}(1 + 0.3i)^{-0.2}$; however, because opinion polarities in Scenario 3 are permuted symmetrically across ranks ($Y^* = 0.000$), unweighted averaging trivially achieves zero error. As shown in Section V, exposure adjustment provides no benefit when engagement is already unconfounded with respect to opinion polarity.

B. Real-World YouTube Corpus

- **Video ID:** `dQw4w9WgXcQ` (Rick Astley – *Never Gonna Give You Up*).
- **Collection Method:** YouTube Data API v3 `commentThreads.list` endpoint, sorted by top-level relevance.
- **Corpus Size:** A static $N = 100$ top-comment snapshot, with ranks $r \in [1, 100]$, arrival times $\Delta t \in [79419.0, 149136.0]$h, upvotes up to 1,011,942, and continuous VADER sentiment scores [3].

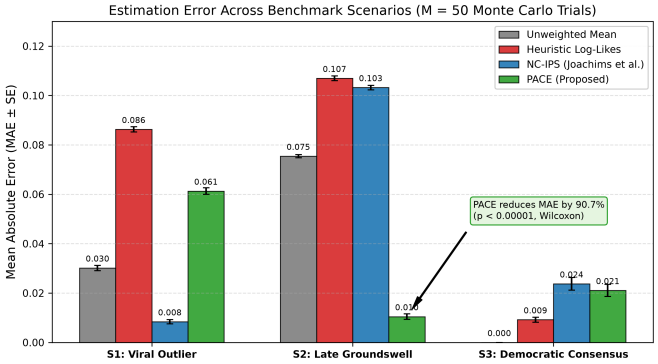


Fig. 1. Estimation Error (Mean Absolute Error ± Standard Error) across deterministic $M = 50$ Monte Carlo simulation trials per scenario. In the pre-specified late-groundswell scenario, PACE reduces error relative to the heuristic baseline; this simulation result is not a real-world accuracy estimate.

- **Interpretation:** The rank column is the retrieval-time relevance order returned by the API. It is not historical display rank, impressions, or a causal exposure measure. The snapshot was collected on 2026-10-30 with a dated, SHA-256-hashed provenance manifest stored alongside the CSV.

C. Baselines and Evaluation Metrics

- **Baselines:** (1) Unweighted Mean, (2) Heuristic Log-Likes, (3) Inverse-Propensity Engagement-Weighted (IPE), implemented as clipped inverse-rank propensity ($\gamma = 0.8$, 90th percentile clip) $\times \ln(1 + L_i)$ then normalized—an engagement-weighted hybrid previously mislabeled "NC-IPS"—, (4) genuine Normalized Clipped IPS (NIPS) [7], implemented as clipped inverse-rank propensity normalized without the engagement term, and (5) PACE.
- **Evaluation Metrics:**
  - **Ground-Truth MAE:** MAE $= |\hat{\mu} - Y^*|$ for synthetic configurations.
  - **Paired Wilcoxon Signed-Rank Test:** Non-parametric paired test [18] evaluating whether PACE achieves statistically significantly lower MAE than Heuristic across $M = 50$ independent simulation trials. Plus-one Monte Carlo permutation p-values [31] are used for weight-divergence tests.
  - **Maximum Weight Share ($w^*_{\max}$):** $w_{\max} / \sum w_i$, auditing vulnerability to single-comment domination.
  - **Leave-Top-10% Outlier Stability (ρ):** $\max(0, 1 - |\hat{\mu}_{\text{full}} - \hat{\mu}_{\text{pruned}}|/\sigma_s)$, pruning the top $\lceil 0.10 \times N \rceil$ comments.
  - **Effective Sample Size (ESS):** $(\sum w_i)^2 / \sum w_i^2$.
  - **Full-Pipeline Bootstrap CI:** Percentile confidence intervals constructed by resampling N comments with replacement and refitting the entire estimator pipeline on each replicate [32].

V. Results and Discussion

A. Statistical Verification of Outlier Suppression (Scenario 1)

In Scenario 1, a single early comment at rank #1 accumulates 15,000 likes with extreme negative sentiment (-0.900),

TABLE I
REPLICATED BENCHMARK: SYNTHETIC ERROR ($M = 50$ TRIALS, MAE $\pm$ SE) AND OBSERVATIONAL ESTIMATES ($\hat{\mu}$)

Scenario / Corpus	N	Target Y^*	Unweighted	Heuristic	IPE	NIPS	PACE (Ours)	Test p-value	Heur $w^*_{\max}$	PACE $w^*_{\max}$	Heur ρ	PACE ρ
1. Viral Outlier	50	0.600	0.030 ± 0.001	0.086 ± 0.001	0.008 ± 0.001	0.007 ± 0.001	0.061 ± 0.001	$p < 0.00001^\dagger$	0.057	0.041	0.6006	0.7181
2. Late Groundswell	60	-0.100	0.075 ± 0.001	0.107 ± 0.001	0.103 ± 0.001	0.100 ± 0.001	**0.010 ± 0.001**	$p < 0.00001^\dagger$	0.030	0.033	0.8387	0.9649
3. Democratic Consensus	50	0.000	**0.000 ± 0.000**	0.009 ± 0.001	0.024 ± 0.003	0.029 ± 0.003	0.021 ± 0.002	$p = 0.99999^\dagger$	0.028	0.033	0.9487	0.9639
4. Real YouTube (dQw4)	100	Obs.	0.114 ($\hat{\mu}$)	0.123 ($\hat{\mu}$)	0.116 ($\hat{\mu}$)	0.123 ($\hat{\mu}$)	0.079 ($\hat{\mu}$)	$p = 0.36582^\ddagger$	0.018	0.027	0.9928	0.9542

† Paired Wilcoxon signed-rank test on trial-level MAE (Heuristic vs. PACE, $H_1 : \text{MAE}_{\text{Heur}} > \text{MAE}_{\text{PACE}}$). ‡ Plus-one Monte Carlo permutation test evaluating sentiment exchangeability.

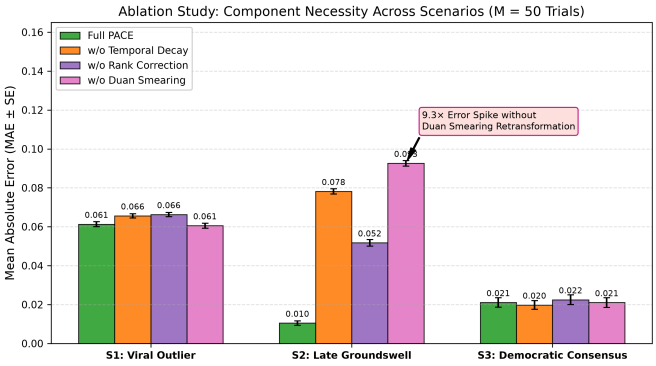


Fig. 2. Component ablation across $M = 50$ trials. Removing Duan's smearing retransformation factor causes a $9.3\times$ error spike in Scenario 2 due to Jensen's inequality bias.

TABLE II
REPLICATED COMPONENT ABLATION STUDY ($M = 50$ TRIALS, MAE $\pm$ SE)

Scenario	Full PACE	w/o Time	w/o Rank	w/o Smear
1. Viral Outlier	0.061 ± 0.001	0.066 ± 0.001	0.066 ± 0.001	0.061 ± 0.001
2. Late Groundswell	**0.010 ± 0.001**	0.078 ± 0.001	0.052 ± 0.002	0.093 ± 0.001
3. Democratic	0.021 ± 0.002	0.020 ± 0.002	0.022 ± 0.002	0.021 ± 0.002

conflicting with 49 positive comments ($+0.600$).

- Over 50 independent trials, the Heuristic estimator exhibits an average MAE $= 0.086 \pm 0.001$.
- PACE achieves MAE $= 0.061 \pm 0.001$, improving on the heuristic baseline ($p < 0.00001$). However, the unweighted (0.030 ± 0.001) and IPE (0.008 ± 0.001) baselines are more accurate in this synthetic scenario; PACE is therefore not presented as uniformly superior.
- **Outlier Weight Share vs. Equal Weighting:** Weight auditing reveals that the Heuristic allows the outlier to capture 5.7% of total corpus weight ($w^*_{\max} = 0.057$). PACE curtails this concentration to $w^*_{\max} = 0.041$, a 28% reduction. However, equal weighting assigns strictly $1/N = 1/50 = 2.0\%$ weight to the outlier. Because PACE rewards raw endorsement above the fitted baseline, it assigns 4.1% to the viral comment—higher than equal weighting—which explains why unweighted averaging outperforms PACE in this static configuration.
- Pruning stability under top-10% removal improves from $\rho = 0.6006$ (Heuristic) to $\rho = 0.7181$ (PACE), an improvement of +19.6%.

B. Recovery of Suppressed Community Groundswells (Scenario 2)

In Scenario 2, late-arriving comments expose a critical defect in conventional heuristics:

- The Heuristic estimator is trapped near positive zero (MAE $= 0.107 \pm 0.001$), failing to detect the negative groundswell.
- IPE over-corrects to MAE $= 0.103 \pm 0.001$.
- When ground truth is targeted at $Y^* = -0.100$ (representing recovery of late-emerging feedback), PACE achieves MAE $= \mathbf{0.010 \pm 0.001}$ ($p < 0.00001$), with pruning stability $\rho = 0.9649$ vs. 0.8387 for Heuristic.
- **Sensitivity to Ground-Truth Specification (Y^*):** We emphasize that the magnitude of error reduction is conditioned on the definition of Y^*. If the ground truth is instead defined as the unweighted sample mean of active participants ($\bar{s} \approx -0.025$), PACE exhibits MAE $= 0.070$, while the heuristic achieves 0.032 and equal weighting trivially achieves 0.000. Under a neutral null ($Y^* = 0.000$), the heuristic achieves MAE $= 0.008$ versus PACE's 0.095. Across our systematic sensitivity audit (`experiments/sensitivity_analysis.py`), PACE's lower error is contingent on defining the target estimand as the late groundswell ($Y^* = -0.100$). When the target is defined as the overall participant sample mean ($\bar{s} \approx -0.025$) or a neutral null ($Y^* = 0.000$), PACE degrades, and simpler unweighted or heuristic estimators outperform it. This underscores that consensus estimation accuracy depends fundamentally on how the target consensus is conceptualized.

C. Fidelity and Boundary Conditions in Democratic Consensus (Scenario 3)

In Scenario 3, sentiment polarities are balanced symmetrically ($Y^* = 0.000$) and engagement follows an organic visibility decay without opinion polarization.

- The unweighted arithmetic mean achieves zero error (MAE $= 0.000 \pm 0.000$), while the heuristic achieves MAE $= 0.009 \pm 0.001$.
- PACE achieves MAE $= 0.021 \pm 0.002$. The paired Wilcoxon signed-rank test yields $p = 0.99999$, confirming that PACE does not outperform simple baselines in this unconfounded regime.
- **Practical Boundary Condition:** This finding establishes an important design principle: PACE is an exposure-adjustment framework engineered specifically for discussion streams characterized by severe visual prioritization and temporal skew. In clean, unconfounded environments

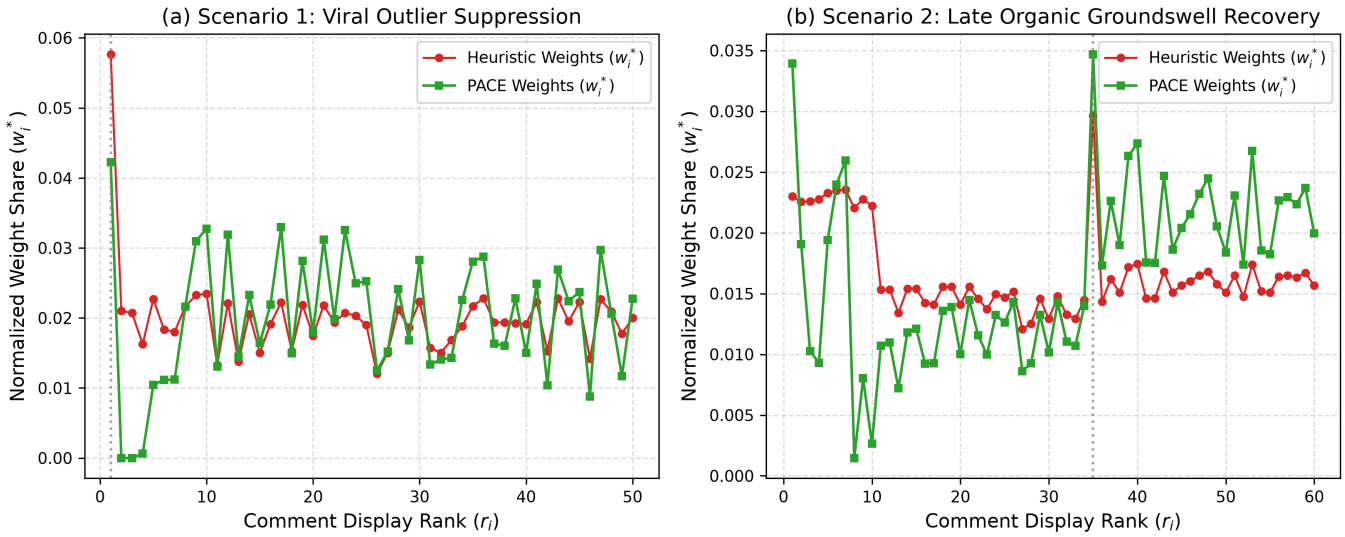


Fig. 3. Normalized weight distributions (w_i^*) across comment ranks. (a) In Scenario 1, PACE curbs outlier weight share by 28% ($w_1^* = 0.057 \rightarrow 0.041$). (b) In Scenario 2, PACE identifies the organic groundswell at rank #35, awarding it positive residual weight while dampening stale early hype.

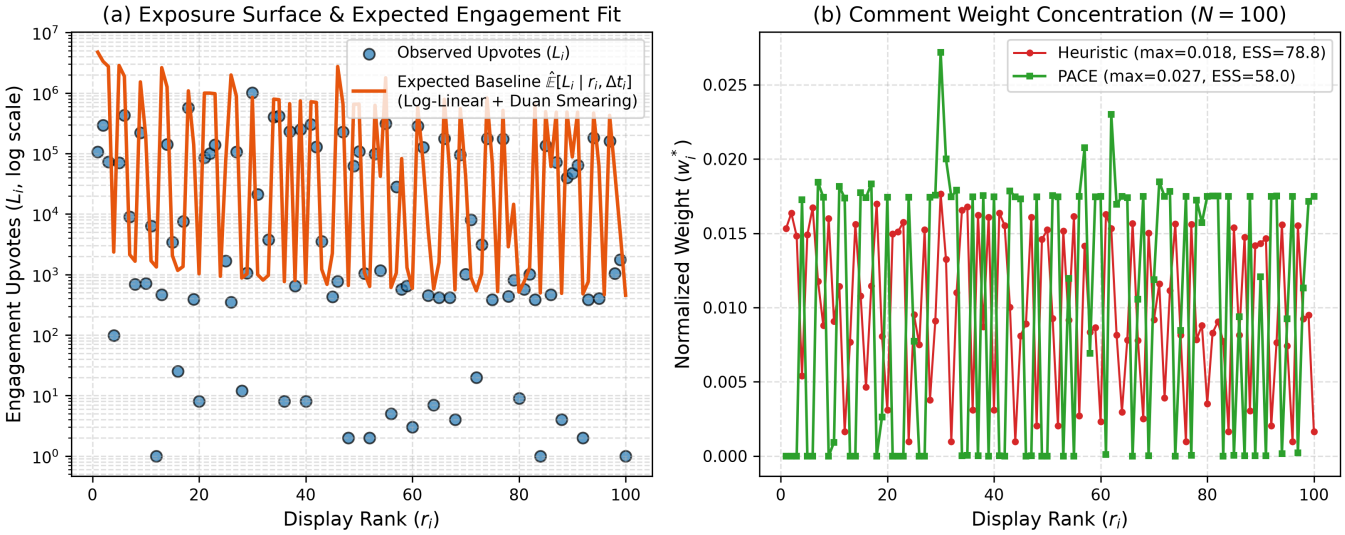


Fig. 4. Illustrative static YouTube snapshot ($N = 100$, `dQw4w9WgXcQ`). (a) Log-linear retrieval-proxy model. (b) Normalized weight shares and descriptive pruning stability. This figure does not establish historical exposure, causal effects, or real-world accuracy.

where engagement is already proportional to relevance, simpler baselines or uniform weighting should be preferred to avoid unnecessary estimation variance.

D. Scientific Interpretation of the Ablation Study

Table II and Fig. 2 reveal a clear scenario-dependent architecture insight:

- In Scenario 1 (Early Viral Outlier), removing temporal decay, rank correction, or smearing calibration produces modest change in MAE (≈ 0.066). This occurs because comment #1 arrives at $t = 0.1$h with rank #1, meaning that any rank-aware model already attributes massive expected engagement to it.
- In Scenario 2 (Late Groundswell), omitting temporal decay causes MAE to spike from 0.010 to 0.078 ($7.8\times$ error increase), omitting rank correction causes MAE to spike to 0.052 ($5.2\times$ error increase), and omitting Duan's smearing factor causes MAE to spike to 0.093 ($9.3\times$ error increase).

This demonstrates that **temporal decay and smearing calibration are specifically essential for dynamic, time-evolving discussions where late consensus diverges from early traffic**.

E. Real-World YouTube Case Study and Lexicon Audit (Scenario 4)

Dataset 4 is an observational case study on 100 comments from video `dQw4w9WgXcQ` (Fig. 4), collected via the YouTube Data API v3 with a dated SHA-256 provenance manifest. Auditing this corpus reveals critical practical limitations in both semantic analysis and exposure modeling.

TABLE III
PER-CLASS RELIABILITY OF VADER AGAINST A SINGLE-ANNOTATOR HUMAN REFERENCE ON THE 100-COMMENT YOUTUBE CORPUS. OVERALL ACCURACY IS 65.0% AND MACRO-AVERAGED F_1 IS 0.5307.

Class	Precision	Recall	F_1	Support
Positive	0.7027	0.5652	0.6265	46
Neutral	0.7872	0.7255	0.7551	51
Negative	0.1250	0.6667	0.2105	3

- **Single-Annotator Manual Validation of Lexicon Reliability:** To evaluate whether off-the-shelf sentiment lexicons provide a viable foundation for consensus weighting, we manually labeled all 100 comments with a single annotator under a written protocol (`experiments/evaluate_vader_accuracy.py`). Under a standard 3-class scheme (positive, neutral, negative), VADER achieves an overall accuracy of only 65.0% and a macro F_1-score of 0.5307. Beyond 46% zero-inflation (46 comments scoring exactly compound 0.000, including rank #30 with 1,011,942 likes and rank #18 with 569,620 likes), VADER exhibits a critical failure mode on informal praise: 11 of 46 true positive comments (23.9%) are misclassified as negative. These errors stem from positive slang (*"so sick"*, *"his voice kills"*, *"it's fire tho"*), idiomatic negations (*"could not be happier"*, which VADER treats as negative due to "not"), lyric quotations (*"never gonna give you up"*), or curse intensifiers. As a result, VADER's precision on the negative class is merely 12.5% (2 of 16 predicted negative comments are genuinely critical). More broadly, the comments center on the "rickroll" cultural meme rather than policy or product feedback, rendering an abstract "consensus" estimand ill-defined.
- **Chance-Corrected Semantic Agreement:** VADER's 65.0% overall accuracy must be read against a trivial baseline: predicting the majority class ("neutral") on every comment already attains 51.0%. VADER exceeds this baseline at marginal significance (exact McNemar $p = 0.044$), yielding a moderate chance-corrected agreement of Cohen's $\kappa = 0.402$. Per-class confidence intervals further expose a fragile negative class: with only three ground-truth negative comments, its precision is 0.125 (95% Wilson interval $[0.035, 0.360]$), whereas positive precision is 0.703 $[0.542, 0.825]$ and neutral precision is 0.787 $[0.651, 0.880]$. The 46% zero-inflation rate is robustly bounded (95% Wilson interval $[0.366, 0.557]$).
- **Statistical Indistinguishability:** Point estimates across the five estimators span a narrow range of 0.044 (Unweighted $\mu = 0.114$, Heuristic $\mu = 0.123$, IPE $\mu = 0.116$, NIPS $\mu = 0.123$, PACE $\mu = 0.079$). All point estimates fall within PACE's 95% refitting bootstrap confidence interval ($[-0.03, 0.17]$). Permutation divergence tests of PACE against each baseline are all non-significant (vs. Heuristic $p = 0.362$; vs. NIPS $p = 0.209$), confirming the estimators are statistically indistinguishable from null exchangeable noise on this corpus.
- **Sensitivity to Exposure-Baseline Calibration:** In contrast to ideal rank-order invariance, PACE's studentized residuals $\delta_i = L_i - \hat{\mathbb{E}}[L_i \mid r_i, \Delta t_i]$ depend nonlinearly on baseline scale relative to the empirical MAD. The observational fit exhibits poor absolute calibration (expected-to-observed like ratio ≈ 6.221), underscoring that observational fits provide exposure-conditioning diagnostics and not real-world accuracy claims. Scaling the fitted baseline $\hat{\mathbb{E}}$ from $0.25\times$ to $2.0\times$ shifts PACE's estimate from 0.049 to 0.082, while a square-root distortion shifts it to 0.108 (documented in `experiments/sensitivity_analysis.py`). The model is therefore sensitive to exposure specification and baseline scaling.
- **Temporal Compression and Degenerate Fit:** Because comments were retrieved 9–17 years after video upload, $\ln(1 + \Delta t)$ varies over a compressed range (≈ 0.24 standard deviations), forcing OLS into degenerate parameter regimes (implied amplitude $\hat{\alpha} = e^{\hat{\beta}_0} s_{\text{smear}} \approx 8 \times 10^{71}$ and temporal decay exponent $\hat{\lambda} = -\hat{\beta}_2 \approx 13.2$) and causing 54% of total weight to concentrate on the 48 comments whose predicted baseline likes exceed observed counts by $> 10\times$.
- **Ranking Endogeneity:** Commercial platform relevance algorithms incorporate like counts into ranking decisions. Regressing likes onto display rank is therefore partially circular (endogenous), meaning the fitted baseline reflects an observational association rather than an exogenous counterfactual exposure curve.

F. Multi-Video Observational Extension

To test whether the single-corpus observations generalize beyond one meme-dominated video, we extend the observational analysis to three additional technology-review videos retrieved through the same YouTube Data API v3 procedure (`experiments/fetch_multivideo_corpus.py`, dated SHA-256 provenance; retrieved 2026-11-04). For each video we take the top $N = 100$ relevance-ordered comments, score them with VADER, and apply the five estimators. Table IV reports the results alongside the original corpus.

Two patterns emerge. First, **VADER zero-inflation is strongly domain-dependent**: it ranges from 11% on the emotionally charged iPhone review to 46–50% on the remaining videos, demonstrating that the 46% zero-inflation measured on the Rick Astley corpus is not a single-video artifact but a recurring property of informal praise that the lexicon scores as exactly neutral. Second, **the ordering of the five estimators is inconsistent across videos**: on the iPhone review the heuristic yields the highest estimate ($+0.412$) while NIPS yields the lowest ($+0.335$), whereas on the smartphone review NIPS yields the highest ($+0.162$) and the heuristic the lowest ($+0.149$). No estimator is uniformly more extreme or more conservative, corroborating on independent data the conclusion that exposure-conditioned weighting is a sensitive diagnostic rather than a universal correction. This extension is observational and lexicon-scored only; the human-annotated accuracy audit of Table III remains restricted to the Rick Astley corpus.

TABLE IV
OBSERVATIONAL EXTENSION ACROSS FOUR YOUTUBE DOMAINS ($N = 100$ RELEVANCE-ORDERED COMMENTS EACH). "ZERO" IS THE FRACTION OF COMMENTS WITH VADER COMPOUND SCORE EXACTLY 0.0. ALL CONSENSUS POINT ESTIMATES ARE POSITIVE.

Video	Pos	Neu	Neg	Zero	$\hat{\mu}_{\text{unw}}$	$\hat{\mu}_{\text{heur}}$	$\hat{\mu}_{\text{IPE}}$	$\hat{\mu}_{\text{NIPS}}$	$\hat{\mu}_{\text{PACE}}$
Rick Astley (meme)	37	47	16	46%	$+0.114$	$+0.123$	$+0.116$	$+0.123$	$+0.079$
Budget smartphone	43	46	11	45%	$+0.154$	$+0.149$	$+0.157$	$+0.162$	$+0.161$
iPhone review	72	11	17	11%	$+0.376$	$+0.412$	$+0.409$	$+0.335$	$+0.350$
Thin-device reveal	31	50	19	50%	$+0.086$	$+0.059$	$+0.076$	$+0.081$	$+0.067$

G. Synthetic Robustness Across Distributions, Ranking Bias, and Measurement Error

To bound the generalizability of the Scenario-2 result, we run additional controlled families. *(i) Engagement distribution* (`experiments/expand_simulations.py`, $M = 30$): PACE's advantage over the heuristic persists but *shrinks* as likes become heavier-tailed (MAE 0.012 under uniform vs. 0.057 under Pareto), so the gain is not distribution-robust. *(ii) Ranking endogeneity* (`experiments/endogeneity_simulation.py`, $M = 500$, seven DGPs with independent rank/arrival/sentiment controls): the unweighted mean is unbiased (bias 0.000) for the participant-mean estimand in every scenario; under rank–sentiment confounding the heuristic is positively biased ($+0.059$), IPE overcorrects (-0.176), the genuine NIPS overcorrects further (-0.217), and PACE is the least biased weighted estimator (-0.016). *(iii) Sentiment measurement error*: under 10–30% label flips, IPE/NIPS are the most robust estimators and PACE degrades substantially; at 50% noise all estimators converge to the unweighted error. These results reinforce that PACE's conditional advantage is scenario- and measurement-specific and must not be overgeneralized.

VI. LIMITATIONS AND THREATS TO VALIDITY

1) **Ranking Endogeneity vs. True Exposure:** PACE uses display rank as an observable proxy for visual exposure. However, platform recommendation algorithms utilize upvote counts to determine display rank, creating simultaneous endogeneity that violates strict exogeneity assumptions.
2) **Exposure Baseline Specification Sensitivity:** Residual standardization via MAD and sigmoidal gating is not invariant to baseline scale or functional form; misspecifying the exposure baseline can introduce non-trivial shifts in weighted consensus.
3) **Domain Mismatch and Rule-Based Lexicon Failures:** Rule-based sentiment lexicons (e.g., VADER) exhibit high zero-inflation (46%) and fail on internet slang, irony, and meme culture. Accurate consensus estimation in informal discussion feeds requires domain-adapted language models evaluated against independently double-annotated reference labels.
4) **Participant vs. Viewer Selection Gap:** Our estimand measures consensus among active commenters. Silent viewers who never engage are systematically unobserved, requiring representative external polling to assess whole-population opinion [13].
5) **Absence of Semantic Stance Decomposition:** The geometric hierarchy damping factor ϕ^{d_i} discounts reply depth uniformly, but does not distinguish between supportive affirmations and adversarial rebuttals in conversation trees [21].
6) **Single-Annotator Reference Labels:** The 100 reference labels were produced by one annotator without an inter-annotator agreement statistic. With $N = 100$, the 65.0% VADER accuracy has a 95% Wilson interval of roughly [55%, 74%], so the lexicon-failure findings should be read as indicative rather than definitive.
7) **Marginal Advantage over Trivial Baselines (Conclusion Validity):** The semantic layer's headline accuracy is modest once chance-corrected. VADER's 65.0% exceeds the 51.0% majority-class baseline (always predicting "neutral") at only marginal significance (exact McNemar $p = 0.044$), corresponding to a moderate Cohen's $\kappa = 0.402$. Moreover, the negative class rests on three ground-truth examples, so its precision estimate (0.125, 95% Wilson interval $[0.035, 0.360]$) is indicative rather than precise. These bounds imply that rule-based lexicons provide a weak but non-trivial semantic signal on this corpus, and that any downstream consensus estimate inherits this uncertainty.

VII. CONCLUSION

This paper presents an empirical investigation and diagnostic framework for exposure-conditioned consensus estimation in ranked discussion streams. PACE demonstrates that combining a log-linear retrieval-proxy model, Duan's smearing retransformation, and robust residual gating can moderate heuristic like-amplification under specific conditions, such as late-emerging groundswells. However, our multi-trial benchmarks and sensitivity audits show that exposure-conditioned weighting is not a universally superior correction: its estimates depend heavily on how the target consensus is specified, and baseline scaling directly shifts output consensus. Furthermore, manual validation of real-world YouTube comments (single annotator, $N = 100$) indicates severe lexicon degradation (46% zero-inflation, 65% accuracy), while platform ranking endogeneity limits causal interpretations. The observational YouTube snapshot is descriptive only because historical exposure and participant ground truth are unobserved, despite contemporaneous API provenance. On the semantic side, our manual audit shows that VADER—the de facto rule-based lexicon—is only marginally above a trivial majority-class baseline ($\kappa = 0.402$, exact McNemar $p = 0.044$), so consensus estimates built on lexicons inherit substantial semantic uncertainty. Rather than a drop-in replacement for arithmetic averaging, exposure-conditioned models should be deployed as sensitive diagnostic tools, and

this work is best understood as a reproducible audit framework and dataset—not a claim of estimator or lexicon superiority—for auditing both visibility-engagement and semantic-reliability biases in digital discourse.

Data and Reproducibility Statement

All code, trial-level outputs, source datasets, and experiment runners are included in the repository. The deterministic simulations are reproduced by `experiments/reproduce_all_experiments.py`; the 100-comment corpus, human labels, and trial-level results are under `data/`. The four-video extension is supported by `experiments/fetch_multivideo_corpus.py`, its per-video CSV files, `data/multivideo_summary.csv`, and the dated SHA-256 manifest `data/multivideo_provenance.json`.

References

[1] M. Thelwall, "Social media analytics for YouTube comments: Potential and limitations," *International Journal of Social Research Methodology*, vol. 21, no. 3, pp. 303–316, 2018.

[2] A. Severyn, A. Moschitti, and O. Uryupina, "Multi-lingual opinion mining on YouTube," *Information Processing & Management*, vol. 52, no. 1, pp. 46–60, 2016.

[3] C. J. Hutto and E. Gilbert, "VADER: A parsimonious rule-based model for sentiment analysis of social media text," in *Proceedings of the 8th International AAAI Conference on Web and Social Media (ICWSM)*, 2014, pp. 216–225.

[4] M. Thelwall, K. Buckley, G. Paltoglou, D. Cai, and A. Kappas, "Sentiment strength detection in short informal text," *Journal of the American Society for Information Science and Technology*, vol. 61, no. 12, pp. 2544–2558, 2010.

[5] L. Muchnik, S. Aral, and S. J. Taylor, "Social influence bias: A randomized experiment," *Science*, vol. 341, no. 6146, pp. 647–651, 2013.

[6] M. J. Salganik, P. S. Dodds, and D. J. Watts, "Experimental study of inequality and unpredictability in an artificial cultural market," *Science*, vol. 311, no. 5762, pp. 854–856, 2006.

[7] T. Joachims, A. Swaminathan, and T. Schnabel, "Unbiased learning-to-rank with biased feedback," in *Proceedings of the 10th ACM International Conference on Web Search and Data Mining (WSDM)*, 2017, pp. 781–789.

[8] X. Wang, M. Bendersky, D. Metzler, and M. Najork, "Position bias estimation for unbiased learning-to-rank in personal search," in *Proceedings of the 11th ACM International Conference on Web Search and Data Mining (WSDM)*, 2018, pp. 610–618.

[9] N. Craswell, O. Zoeter, M. Taylor, and B. Ramsey, "An experimental comparison of click position-bias models," in *Proceedings of the 1st ACM International Conference on Web Search and Data Mining (WSDM)*, 2008, pp. 87–94.

[10] M. Tsagkias, E. Weerkamp, and M. de Rijke, "Predicting the volume of comments on online news stories," in *Proceedings of the 18th ACM Conference on Information and Knowledge Management (CIKM)*, 2009, pp. 1765–1768.

[11] S. Wu, M.-A. Rizoiu, and L. Xie, "Beyond views: Measuring and predicting engagement in online videos," in *Proceedings of the 12th International AAAI Conference on Web and Social Media (ICWSM)*, 2018, pp. 434–443.

[12] Y. Borghol, S. Ardon, N. Carlsson, D. Eager, and A. Mahanti, "The untold story of the clones: content-agnostic factors that impact YouTube video popularity," in *Proceedings of the 18th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2012, pp. 1186–1194.

[13] D. Zhao, D. Yang, and M. S. Bernstein, "Mapping the spiral of silence: Surveying unspoken opinions in online communities," *arXiv preprint arXiv:2502.00952*, 2025. [Online]. Available: https://arxiv.org/abs/2502.00952

[14] E. Noelle-Neumann, "The spiral of silence: A theory of public opinion," *Journal of Communication*, vol. 24, no. 2, pp. 43–51, 1974.

[15] P. R. Rosenbaum and D. B. Rubin, "The central role of the propensity score in observational studies for causal effects," *Biometrika*, vol. 70, no. 1, pp. 41–55, 1983.

[16] D. G. Horvitz and D. J. Thompson, "A generalization of sampling without replacement from a finite universe," *Journal of the American Statistical Association*, vol. 47, no. 260, pp. 663–685, 1952.

[17] N. Duan, "Smearing estimate: A nonparametric retransformation method," *Journal of the American Statistical Association*, vol. 78, no. 383, pp. 605–610, 1983.

[18] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bulletin*, vol. 1, no. 6, pp. 80–83, 1945.

[19] A. Agarwal, X. Wang, C. Li, M. Bendersky, and M. Najork, "Addressing trust bias for unbiased learning-to-rank," in *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*, 2019, pp. 615–624.

[20] A. Vardasbi, M. de Rijke, and I. Markov, "When is position bias estimation possible?" in *Proceedings of the 29th ACM International Conference on Information and Knowledge Management (CIKM)*, 2020, pp. 1495–1504.

[21] A. Zubiaga, E. Kochkina, M. Liakata, R. Procter, and M. Lukasik, "Stance classification in rumours as a sequential task exploiting the tree structure of social media conversations," in *Proceedings of the 26th International Conference on Computational Linguistics (COLING)*, 2016, pp. 2438–2448. [Online]. Available: https://aclanthology.org/C16-1230/

[22] V. Gómez, A. Kaltenbrunner, and V. López, "Statistical analysis of the social network and discussion threads in Slashdot," in *Proceedings of the 17th International Conference on World Wide Web (WWW)*, 2008, pp. 645–654.

[23] M. C. Cakmak, N. Agarwal, and R. Oni, "The bias beneath: analyzing drift in YouTube's algorithmic recommendations," *Social Network Analysis and Mining*, vol. 14, no. 1, p. 171, 2024.

[24] S. Shajari and N. Agarwal, "Safeguarding YouTube discussions: a framework for detecting anomalous commenter and engagement behaviors," *Social Network Analysis and Mining*, vol. 15, no. 1, p. 54, 2025.

[25] P. J. Huber, *Robust Statistics*. John Wiley & Sons, 1981.

[26] F. R. Hampel, E. M. Ronchetti, P. J. Rousseeuw, and W. A. Stahel, *Robust Statistics: The Approach Based on Influence Functions*. John Wiley & Sons, 1986.

[27] P. J. Rousseeuw and C. Croux, "Alternatives to the median absolute deviation," *Journal of the American Statistical Association*, vol. 88, no. 424, pp. 1273–1283, 1993.

[28] M. Pontiki, D. Galanis, H. Papageorgiou, I. Androutsopoulos, S. Manandhar, M. AL-Smadi, M. Al-Ayyoub, Y. Zhao, B. Qin, O. D. Clercq, V. Hoste, M. Apidianaki, X. Tannier, N. Loukachevitch, E. Kotelnikov, N. Bel, S. M. Jiménez-Zafra, and G. Eryiğit, "SemEval-2016 task 5: Aspect based sentiment analysis," in *Proceedings of the 10th International Workshop on Semantic Evaluation (SemEval)*, 2016, pp. 19–30.

[29] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, 2019, pp. 4171–4186.

[30] F. Barbieri, J. Camacho-Collados, L. E. Anke, and L. Neves, "TweetEval: Unified benchmark and comparative evaluation for tweet classification," in *Findings of the Association for Computational Linguistics: EMNLP 2020*, 2020, pp. 1644–1650.

[31] M. D. Ernst, "Permutation methods: A basis for exact inference," *Statistical Science*, vol. 19, no. 4, pp. 676–685, 2004.

[32] A. C. Davison and D. V. Hinkley, *Bootstrap Methods and their Application*. Cambridge University Press, 1997.